

Multiple Choice (10 marks)

1. What is the percentage of angular magnification when one views an object at 33 cm through a pane of glass 5 mm thick?
 - (a) 0.7%
 - (b) 3%
 - (c) 0.1%
 - (d) 1.5%
 - (e) 1%
2. Calculate the de Broglie Wavelength of a tennis ball of mass 57 g traveling 25 m/s in meters.
 - (a) 3.0×10^{-32}
 - (b) 5.5×10^{-33}
 - (c) 1.2×10^{-30}
 - (d) 8.9×10^{-36}
 - (e) 2.3×10^{-30}
3. A₃₂₀₀-lb car traveling with a speed of 60 mi/hr rounds a curve whose radius is 484 ft. Find the necessary centripetal force.
 - (a) 1200 lb
 - (b) 1800 lb
 - (c) 2000 lb
 - (d) 1000 lb
 - (e) 2400 lb
4. A barometer that uses water instead of mercury will be
 - (a) more fragile
 - (b) shorter
 - (c) heavier
 - (d) equal in height
 - (e) taller
5. Kirkwood gaps are observed in the main asteroid belt including at the position(s) where:

- (a) asteroids would orbit with a period that is three times that of Jupiter
- (b) asteroids would orbit with a period three times that of Mars
- (c) asteroids would orbit with a period twice that of Mars
- (d) asteroids would orbit with a period twice that of Jupiter
- (e) A and B

True / False (10 marks)

Answer True or False

6. True or False: The displacement of hiker 2 is 18 km, which is half the total distance covered by both hikers.
7. True or False: One astronomical unit (AU) is approximately 170 million kilometers, which is often confused with its actual value.
8. True or False: Knowing both the actual brightness and the apparent brightness of an object allows for the estimation of its distance from the observer.
9. True or False: When white light passes through a prism, blue light bends more than green light due to its shorter wavelength.
10. True or False: An object falling from rest will cover a distance of 2 h during the second second of its fall, given the distance fallen in the first second is h.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. A proton, initially at rest far from Earth, falls under gravity and converts its kinetic energy into a quantum of light. Calculate the frequency of this quantum and discuss the underlying principles involved.
12. Discuss the calculation of the frequency for the hydrogen transition from $n = 101$ to $n = 100$. Explain the underlying principles and significance of this transition within atomic physics.

End of Paper